

Study Notes: Numbers & Quantifications

1. Modulo Arithmetic ("Wrap Around" Math)

- Operator: $a \bmod b = n$
 - a = Dividend
 - b = Divisor
 - n = Remainder

The Golden Rule

- Remainder can never be negative and must be smaller than the divisor.

Example: $50 \bmod 12$.

→ $12 \times 4 = 48$. Remainder is 2.
• So, $50 \bmod 12 = 2$.

4. Alligation & Mixture

Cheaper (c) — Mean (m) — Dearer (d)



$d - m : m - c$

Example: Ratio = 1 : 2.

Examples:

- $45 \bmod 7 = 3$
- $-8 \bmod 5 = 2$
- Mix ₹40/kg & ₹70/kg → Ratio = 1 : 2

2. Congruence Modulo

- Symbol: $a \equiv b \pmod{n}$
- Divide both a & b by n .
- OR: $(a - b)$ is a multiple of n .

Example: $12 \equiv 27 \pmod{5}$?

$$27 - 12 = 15$$

Yes, they're congruent!

3. Properties for Fast Calculation

- Addition — $(a + b) \bmod n$
- Multiplication — $(a \times b) \bmod n$
- Exponents (Power) : $a^k \equiv b^k \pmod{n}$

5. Repeated Dilution

- Leftover = $x(1 - y)^n$
 - x = Original quantity.
 - y = Quantity replaced,
 - n = Number of times.

6. Equivalence Class

Format: $[a]_n = \{\dots a - 2n, a, a + n, a + 2n, \dots\}$

UNIT 2: ALGEBRA

Applied Mathematics (Class 12)

1. Matrices

Definition =

- A matrix is a rectangular arrangement of numbers in rows and columns,

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

→ Order of matrix = Rows × Columns

★ Example: 2 rows & 3 columns → 2×3 matrix

Types of Matrices

Row Matrix	Only one row
Column Matrix	Only one column
Square Matrix	Rows = Columns
Zero Matrix	All elements are 0
Identity Matrix (I)	Diagonal elements = 1
Diagonal Matrix	Non-diagonal elements = 0

Equality of Matrices

- Same order
- Corresponding elements equal

3. Inverse of Matrix

★ Formula (2×2 Matrix)

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

4. Solving Simultaneous Equations

★ Matrix form:

$$AX = B$$

$$\star X = A^{-1} B$$

Equality of Matrices

- Same order
- Corresponding elements equal

Operations on Matrices

✓ Addition / Subtraction

$$A + B = [a_{ij} + b_{ij}]$$

✓ Scalar Multiplication

$$kA$$

✓ Matrix Multiplication

$$\text{Columns of } A = \frac{R^U}{B =}$$

$$\star AB \neq BA$$

2. Determinants (2×2)

Determinant Meaning

★ For matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$|A| = ad - bc$$

✓ Properties

- Interchanging rows changes sign.
- If two rows equal → Det = 0.
- Determinant of identity matrix = 1.

Examples

- If $A = \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}$, find $|A|$?
- Identify matrix multiplication.
- Check orders before operations
- Don't forget signs!



UNIT 3 — CALCULUS

(Applied Maths Class 12)



★ 1. Differentiation & Its Applications

○ Meaning of Differentiation

Differentiation measures the rate of change of one quantity with respect to another.

→ If $y = f(x)$, then derivative: $\frac{dy}{dx} = f'(x)$

- Slope of tangent
- Instantaneous rate of change

★ Basic Derivative Formulas ★

Function	Derivative
—	—

○ Rules of Differentiation

✓ **Sum Rule:** $\frac{d}{dx}(u+v) = u' + v'$

✓ **Product Rule:** $(uv)' = u'v + uv'$

✓ **Quotient Rule:** $\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$

✓ **Chain Rule:** $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

★ Applications of Derivatives ★

★ Increasing ↓ / Decreasing ↑

★ Maxima & Minima:

1. Find $f'(x) = 0$
2. Put $f'(x) > 0 \rightarrow$ Minimum
3. Put $f'(x) < 0 \rightarrow$ Maximum

2. Integral Calculus

→ **Meaning:**
Integration is the reverse process of differentiation.

$$\int f(x)dx = F(x) + C$$

○ Standard Integrals

✓ **Substitution Method:**

$$\frac{dy}{dx} = xy \Rightarrow \ln y = \frac{x^2}{2} + C$$

✓ **Integration by Parts:** $\int u \cdot dv = uv - \int v \cdot du = f(x)g(y) \Rightarrow \frac{1}{4}(x^2+1)^2 + C$

3. Differential Equations

→ **Meaning:**

An equation involving derivatives.

Example: $\frac{dy}{dx} = 3x$

— Created By Thunderstudy —

— All topics as given —

UNIT 4: PROBABILITY DISTRIBUTIONS

Class 12 Applied Mathematics

1. Probability Distribution

Meaning

- A Probability Distribution shows how probabilities are assigned to different values of a random variable.
- It tells how likely each outcome is.

Random Variable

- A random variable (X) is a numerical value determined by chance.
- Example: Number of heads when tossing 3 coins.

Types of Random Variables

1. Discrete Random Variable

- Takes countable values.

2. Continuous Random Variable

- Takes any value in an interval.



Conditions of Probability Distribution :

- $\sum P(x) = 1$
- $P(x) \geq 0$

Mean (Expected Value): $E(X) = \sum xP(x)$

Variance: $Var(X) = \sum x^2P(x) - [E(X)]^2$

Standard Deviation: $\sigma = \sqrt{Var(X)}$

Conditions (Very Important)

- n = number of trials
- Outcomes: Success or Failure
- Trials independent
- p = Probability of success
- $q = 1 - p$

Formula:

- $P(X = x) = {}^nC_x p^x q^{n-x}$
- Where:
 - n = number of trials
 - 0 : Only two outcomes (Success/Failure)
 - P Probability remains constant.

2. Binomial Distribution

Meaning

- Fixed number of trials
- Only two outcomes (Success/Failure)
- Probability remains constant.



$$P(X = x) = {}^nC_x p^x q^{n-x}$$

Mean: $\mu = np$

Variance: $\sigma^2 = npq$

$\sigma = \sqrt{npq}$

Example: Probability of getting 2 heads in 3 coin tosses.

Quick Comparison (Exam Favourite)

Distribution	Type	Mean	Variance	Use
Probability Distribution	Discrete	$\sum xP(x)$	Formula based	General cases
Binomial	Discrete	np	npq	Fixed trials
Poisson	Discrete	λ	λ	Rare events
Normal	Continuous	μ	σ^2	Natural data

4. Normal Distribution

Meaning

- Bell-shaped curve
- Natural measurements.
- Examples:
 - Height, Marks, IQ scores.



Properties

- ✓ Symmetrical curve
- ✓ Mean = Median = Mode
- ✓ Total area = 1.

Examples:

- Rolling a die
- Defective items
- Calls per hour
- Heights of students

Unit 5

Inferential Statistics

= Chapter: Primary Activities =

★ 1. Meaning of Inferential Statistics :

- Used to draw conclusions about a population from a sample.
- Study a small part (sample) & make predictions.



Example: Survey 100 students to predict study habits of all.

★ 2. Population and Sample =

Term	Meaning
Population	→ Entire group under study
Sample	→ Small part taken from population
Parameter	→ Numerical measure of population
Statistic	→ Numerical measure of sample

★ 3. Sampling =

• Random Sampling

• Non-Random Sampling

★ 4. Sampling Distribution =

$$\rightarrow SE = \frac{\sigma}{\sqrt{n}}$$

★ 5. Central Limit Theorem (CLT) =

★ 6. Estimation =

⇒ Point & Interval Estimates

★ 11. Type I and Type II Errors =

- Type I: False Positive
- Type II: False Negative

★ 12. Test Statistic (Z-Test) =

$$\rightarrow Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

★ 13. Decision Rule =

- ★ If Z in critical region → Reject H_0
- ★ Else → Accept H_0

★ 3. Sampling =

• Random Sampling

• Non-Random Sampling

★ Central Limit Theorem (CLT) =

- For large samples ($n \geq 30$) → Mean follows normal distribution.

★ 6. Estimation =

→ Point & Interval Estimates

★ 7. Confidence Interval =

★ 8. Hypothesis Testing =

→ Steps:

1. State Hypothesis
2. Set Significance Level
3. Calculate Test Statistic
4. Compare & Decide

★ 9. Types of Hypothesis =

- H_0 : Null Hypothesis
- H_1 : Alternative Hypothesis

★ 10. Level of Significance (α) =

$$\alpha = 0.05 \text{ or } 0.01$$

★ Exam Tips (Very Important) ★ ★

✓ Write:

- ✓ H_0 & H_1
- ✓ Formulas & Steps
- ✓ Clear Conclusion.

UNIT 6 : TIME BASED DATA

— (Applied Mathematics – Class 12) —

1. Meaning of Time Based Data

Time Based Data means data collected over different periods of time to study changes or trends.

♥ Growth. → Decline. → Seasonal patterns. → Future predictions

2. Time Series:

A Time Series is a set of observations arranged in chronological order (according to time)

★ Formula: $Y = f(T)$

♥ Y = Value of variable

♥ T = Time

Year	Sales
2020	100
2021	120
2022	150

3. Components of Time Series

1. Trend (T) – Population growth over years
2. Seasonal Variation (S) – Ice cream sales
3. Cyclical Variation (C) – Economic boom & recession
4. Irregular Variation (I) – Pandemic, natural disaster.

4. Trend Analysis

1. Graphic Method
2. Semi-Average Method
3. Moving Average Method
4. Least Square Method

5. Moving Average Method.

1. Select period
2. Calculate average
3. Move forwards
4. Repeat steps

6. Semi-Average Method.

1. Split the data
2. Find averages
3. Plot & join points

Examples :

- ♥ Montilig Sales
- ♥ Annual Rainfall
- ♥ Daily Tempessions
- ♥ Stock Prices

- ★ Monthly Sales
- ★ Annual Rainfall
- ★ Daily Temperature
- ★ Stock Prices

— Created By Thunderstudy —

Chapter Primary Activities

(Applied Mathematics - Class 12)



1. PERPETUITY

- ✓ **Meaning:** A Perpetuity is a stream of equal payments made forever (infinite time period).

➔ **Example:** Scholarship paid every year forever.



Formula: $PV = \frac{R}{i}$



Perpetuity Due: $PV = \frac{R}{i} (1+i)$



2. SINKING FUND

- ✓ **Meaning:** A Sinking Fund is money saved regularly to repay a future debt or replace an asset.



Formula: $R = \frac{A \times i}{(1+i)^n - 1}$



3. EMI (Equated Monthly Instalment)

- ✓ **Meaning:** EMI is a fixed monthly payment made to repay a loan.



Formula: $EMI = \frac{P \times r \times (1+r)^n}{(1+i)^n - 1}$



4. EMI (Equated Monthly Instalment)

- ✓ **Meaning:** EMI is a fixed monthly payment made to repay a loan.



EMI Formula: $EMI = \frac{P \times r \times (1+r)^n}{(1+i)^n - 1}$



6. DEPRECIATION

- ✓ **Meaning:** Decrease in asset value over time.



4. RETURNS

- ✓ **Meaning:** Return is the profit or loss earned on investment.



Formula: $\text{Return} = \frac{\text{Income}}{\text{Investment}} \times 100$



5. GROWTH

- ✓ **Meaning:** Increase in value over time due to compound interest.



QUICK FORMULA REVISION TABLE

- 1 Perpetuity: $PV = \frac{R}{i}$
- 1 Sinking Fund: $R = \frac{A \times i}{(1+i)^n - 1}$
- 2 EMI: $= \frac{P \times r \times (1+r)^n}{(1+i)^n - 1}$
- 2 Growth: $A = P(1+r)^n$
- 1 Depreciation: $V = P(1-r)^n$
- 2 Return: $\text{Return} = \frac{\text{Income}}{\text{Investment}} \times 100$

★ UNIT 8: LINEAR PROGRAMMING ★

Applied Mathematics – Class 12 Notes

1. Introduction to Linear Programming (LP)

★ Mathematical method to find the best solution (maximum profit or minimum cost) :

- ★ Business planning
- ★ Production management
- ★ Resource allocation
- ★ Transportation problems

Main idea:

- ★ Optimize (Maximize or Minimize) a linear function subject to constraints.

2. Basic Terminology:

1. Decision Variables: = Values to determine.

- ★ Number of products = x, y

3. Constraints: : $Z = 3x + y \leq 10$,

- ★ Restrictions as inequalities. e.g, $2x + y \leq 10$

2. Objective Function:

$$Z = 5x + 3y$$

(Z = Profit / Cost).

3. General Form of LPP

Maximize or Minimize

$$Z = ax + by$$

Subject to:

- ✓ $a_1x + b_1y \leq / \geq c_1$
- ✓ $a_2x + b_2y \leq / \geq c_2$
- ✓ $x, y \geq 0$

4. Non-Negativity Constraints

- ✓ Optimal Solution.
- ✓ Feasible Solution.
- ✓ Infeasible Solution.
- ✓ Multiple Solution:
- ✓ Unbounded Solution.

Examples:

✓ Maximize: $Z = 3x + 2y$

✓ Subject to:

$$x + y \leq 4$$

$$x \leq 2, y \leq 3$$

✓ $x, y \geq 0$.

→ Draw graph & find optimum!

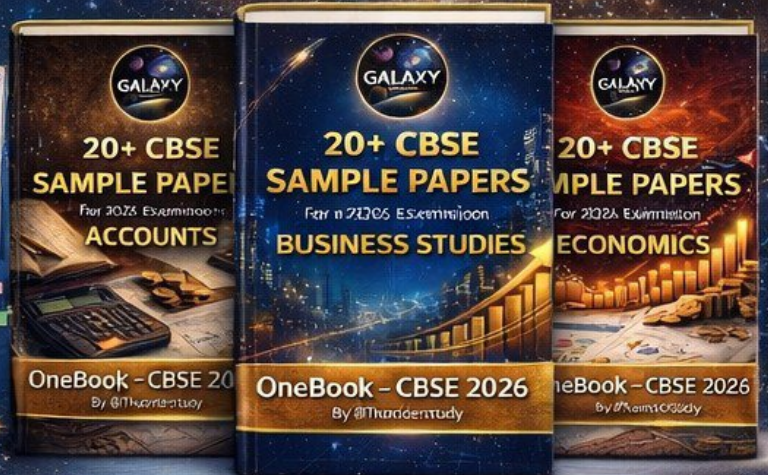
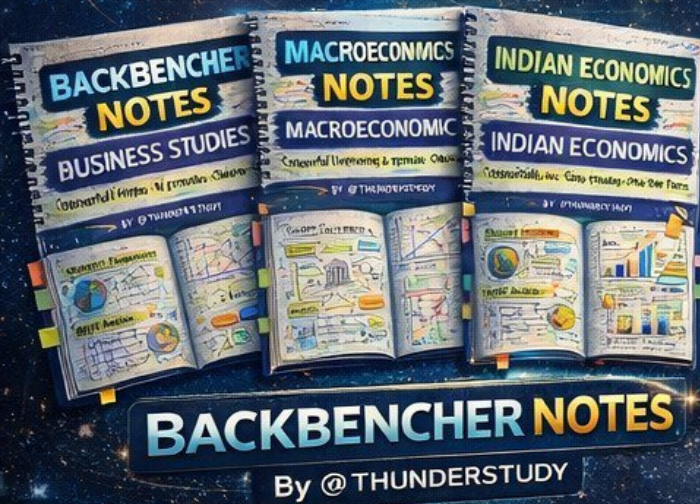
Quick Revision Formula Box:

★ Objective Function	✓ $Z = ax + by$
Constraints	✓ $a_1x + b_1y \leq / \geq c_1$
Solution	✓ Corner Points
Method	✓ Graphical Method

Created By Thunderstudy

By @THUNDERSTUDY

Other By @Thunderstudy



Best Notes Best Book
For Commerce

Scan the QR or
click here to join >

